

Statistical Models in Talos

We can expand on the current Talos by adding the ability to incorporate statistical models that we call from within the Lua script. The idea is simple: instead of writing a mortality model in Lua like this:

```
if person.age < 30 then
  prob = 0.001
elseif person.age >= 30 and person.age < 60 then
  prob = 0.05
else
  prob = 0.10
end
```

You could instead call a proper statistical model trained on real data:

```
local features = { age = person.age, sex = person.sex, smoking = person.
smoking }
local prob = talos_stats.predict("mortality_model", features)
```

The statistical model itself would live in Go, not Lua. We would register a set of Go functions with the Lua interpreter so they can be called just like any other Lua function. The coefficients for these models would be defined in YAML, so researchers can update them without touching any code.

For example, a logistic regression model for mortality might look like this in YAML:

```
statistical_models:
- name: "mortality_model"
  type: "logistic"
  intercept: -4.5
  coefficients:
    age: 0.08
    sex: -0.3
    smoking: 0.5
```

The Go implementation would be completely dynamic - it would take whatever features the Lua script passes and apply the coefficients from the YAML file. There would be no

hardcoded feature names or model types in the Go code. This means adding a new predictor variable to a model requires no changes to the Go code whatsoever. Researchers simply add the column to their CSV, update the coefficients in YAML, and their Lua script automatically passes the new feature to the model.

This approach maintains Talos's core virtues. The system remains a single binary with zero dependencies, as all statistical functionality is built into the Go binary and exposed through the Lua API. Researchers can train models in R or Python, export the coefficients, and use them directly in Talos without modifying the source code. Different models can be applied to different population subgroups, all controlled through YAML configuration. The whole system stays flexible, accessible, and easy to deploy, while gaining the power of proper statistical modeling.